Code No: 131AG **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech I Year I Semester Examinations, March/April - 2023
ENGINEERING CHEMISTRY
(Common to EEE, ECE, CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	Define Hardness and write its different units.	[2]
b)	Discuss advantages of disinfection of potable water by ozone.	[3]
c)	What is the significance of electrochemical series? State it's applications.	[2]
d)	Write chemical reactions involved in Ni-Cd cell during charging.	[3]
e)	Define degree of polymerisation and functionality.	[2]
f)	Write the synthesis of Nylon-6, 6.	[3]
g)	What is octane and cetane number?	[2]
h)	How to remove water and sulphur from crude oil.	[3]
i)	Define Refractoriness.	[2]
j)	Give classification of composites.	[3]

PART - B

(50 Marks)

2.a) What is desalination? Describe the Reverse Osmosis process for softening of hard water.
b) How to estimate fluorine in water by ion-selective method. [5+5]

OR

3.a) Calculate the temporary, permanent and total hardness of the water, containing $MgCl_2$=9.5 mg/L; $Ca(HCO_3)_2$=16.2 mg/L and $CaSO_4$=13.6 mg/L.
b) What are the specifications of portable water? Explain sedimentation with coagulation using suitable equations. [5+5]

4.a) Define fuel cells and describe the construction and working of Methanol-O_2 fuel cell.
b) What is a battery? Explain the construction and functioning of dry cell. [5+5]

OR

5.a) Explain the working principle of glass electrode.
b) What is Concentration cell? Explain with example. [5+5]

6.a) Define polymer and write the free radical addition polymerization mechanism of Olefins.
b) Explain the importance of compounding of plastics. [5+5]

OR

7.a) Give the classification of polymers with suitable examples.
b) Discuss the synthesis, properties and applications of Fibre reinforced plastics. [5+5]

8.a) Calculate the gross and net Calorific value of coal sample having the following composition. C=83%, H=6%, O=3.5%, S=3.5%, N=3.5% and ash 4.0%.
b) How is carbon and hydrogen determined in a solid fuel? [5+5]

OR

9.a) Give the composition and uses of natural gas, LPG and CNG.
b) What is cracking? Explain moving bed catalytic cracking process. [5+5]

10.a) What are the characterstics of good lubricants and give their significance.
b) What is Portland Cement? Write the chemical reactions involved in setting and hardening of Cement. [5+5]

OR

11.a) What are lubricants? Discuss mechanism of Thin film Lubrication.
b) Discuss the differences between the acidic, basic and neutral refractories with suitable examples. [5+5]

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